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United States Patent Application Publication	20250281005
Kind Code	A1
Publication Date	September 11, 2025
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TOWEL HEATING DEVICE

Abstract

A towel heating device includes two body members, a foldable member, and a heating assembly. The two body members are arranged opposite to each other and capable of moving close to or away from each other. The foldable member is connected to the two body members. The foldable member and the two body members cooperatively define a heating cavity. The foldable member folds as the two body members move close to each other and unfolds as the two body members move away from each other. The heating assembly is distributed on the foldable member and at least one of the two body members and configured to heat the heating cavity.

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Family ID:	1000008643256
Appl. No.:	19/216795
Filed:	May 23, 2025

Foreign Application Priority Data

CN	202410260988.9	Mar. 07, 2024
CN	202420441375.0	Mar. 07, 2024
CN	202520812659.0	Apr. 25, 2025

Related U.S. Application Data

parent US continuation-in-part 18763632 20240703 parent-grant-document US 12329326 child US 19216795
parent US continuation-in-part 18763734 20240703 PENDING child US 19216795

Publication Classification

Int. Cl.: A47K10/02 (20060101); F26B9/00 (20060101); F26B23/06 (20060101)

U.S. Cl.:

CPC A47K10/025 (20130101); F26B9/003 (20130101); F26B23/06 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation-in-part application of: the U.S. patent application Ser. No. 18/763,734, filed on Jul. 3, 2024; and the U.S. patent application Ser. No. 18/763,632, filed on Jul. 3, 2024. The present application claims the priority of: the Chinese patent application No. 202410260988.9, filed on Mar. 7, 2024; the Chinese patent application No. 202420441375.0, filed on Mar. 7, 2024; and the Chinese patent application No. 202520812659.0, filed on Apr. 25, 2025. Contents of which are incorporated herein by their entireties.

FIELD

[0002] The present disclosure relates to the field of bathroom appliances, and in particular to a towel heating device.

BACKGROUND

[0003] Towel heating barrels in a related art require heating elements to be mounted in a barrel body to heat a heating cavity in the barrel body. For foldable towel heating barrels, heating elements are typically only mounted on rigid shells on both sides of the foldable towel heating barrels, resulting in limited heating area and low heating efficiency for towels.

SUMMARY

[0004] A towel heating device is provided in the present disclosure. The towel heating device includes two body members, a foldable member, and a heating assembly. The two body members are arranged opposite to each other and capable of moving close to or away from each other. The foldable member is connected to the two body members. The foldable member and the two body members cooperatively define a heating cavity. The foldable member folds as the two body members move close to each other and unfolds as the two body members move away from each other. The heating assembly is distributed on the foldable member and at least one of the two body members and configured to heat the heating cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] To provide a clearer illustration of the technical solutions in the embodiments of the present disclosure or in the related art, a brief introduction will be given to the drawings used in the description of the embodiments or the related art. It is obvious that the drawings described below are merely some embodiments of the present disclosure, and for those skilled in the art, other drawings can be obtained based on these drawings without creative efforts.

[0006] FIG. 1 is a schematic view of an overall structure of a towel heating device according to an embodiment of the present disclosure.

[0007] FIG. 2 is a structural view of a foldable member connecting two body members according to an embodiment of the present disclosure.

[0008] FIG. 3 is an exploded view of the foldable member and the two body members according to an embodiment of the present disclosure.

[0009] FIG. 4 is a structural view of four side walls and one bottom wall corresponding to a

mounting cavity according to an embodiment of the present disclosure.

[0010] FIG. 5 is a cross-sectional view of a partial structure of the foldable member according to an embodiment of the present disclosure.

[0011] FIG. 6 is a first structural view of a flexible heating sheet according to an embodiment of the present disclosure.

[0012] FIG. 7 is a second structural view of the flexible heating sheet according to an embodiment of the present disclosure.

[0013] FIG. 8 is a structural view of the towel heating device according to an embodiment of the present disclosure, showing a bottom portion of the towel heating device.

[0014] FIG. 9 is an enlarged view of a portion A in FIG. 8.

[0015] FIG. 10 is a cross-sectional view of a partial structure of the towel heating device according to an embodiment of the present disclosure.

[0016] FIG. 11 is an exploded view of a body member according to an embodiment of the present disclosure.

[0017] FIG. 12 is cross-sectional view of a partial structure of a heating plate according to an embodiment of the present disclosure.

REFERENCE NUMERALS IN THE DRAWINGS

[0018] **1**—body member; **10**—mounting cavity; **11**—shell; **12**—mounting frame; **111**—bottom portion of the shell; **112**—first wire hole; **113**—second wire hole; **114**—wire winding structure; **121**—bottom portion of the mounting frame; **122**—thermal insulation net; [0019] **2**—foldable member; **20**—partition; **21**—flexible outer layer; [0020] **3**—heating assembly; **31**—control board; **32**—flexible heating sheet; **33**—heating plate; **321**—first mounting sheet; **322**—first heating wire; **331**—second mounting sheet; **332**—second heating wire; **333**—metal plate; **3211**—first heat conduction layer; **3212**—thermal insulation layer; **3220**—heating portion; **3221**—connection end; **3311**—second heat conduction layer; **3312**—third heat conduction layer; [0021] **4**—power cord; [0022] **100**—heating cavity; **1001**—side wall; **1002**—bottom wall.

DETAILED DESCRIPTION

[0023] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art to which the present disclosure belongs. The terms used in the description of the present disclosure herein are intended for describing particular embodiments only and are not intended to limit the present disclosure. In the description, claims, and the above drawings of the present disclosure, the terms “including” and “having”, as well as their variants, are intended to convey a non-exclusive inclusion. The terms “first”, “second”, etc., as used herein, are intended to distinguish between different objects, rather than to describe a particular order.

[0024] Reference to “embodiments” herein implies that a particular feature, structure, or characteristic described in conjunction with an embodiment may be included in at least one embodiment of the present disclosure. The appearance of the phrase at various places in the specification does not necessarily refer to the same embodiment, nor is it a separate or an alternative embodiment that is mutually exclusive of other embodiments. One skilled in the art would explicitly and implicitly understand that the embodiments described herein can be combined with other embodiments.

[0025] In order to enable those skilled in the art to better understand the technical solutions of the present disclosure, the technical solutions in the embodiments of the present disclosure will be clearly and completely described below with reference to the accompanying drawings.

[0026] Referring to FIGS. 1 to 12, the present disclosure provides a towel heating device. The towel heating device includes two body members **1**, a foldable member **2**, and a heating assembly **3**. The two body members **1** are arranged opposite to each other and are capable of moving close to or away from each other. Each of the two body members **1** may be a rigid structure, a flexible structure, or one of the two body members **1** may be a rigid structure, while the other of the two

body members **1** may be a flexible structure, depending on usage requirements.

[0027] The foldable member **2** is connected to the two body members **1**. The foldable member **2** and the two body members **1** cooperatively define a heating cavity **100**. The foldable member **2** folds as the two body members **1** move close to each other and unfolds as the two body members **1** move away from each other. In one embodiment, the foldable member **2** is composed of a plurality of hard plates that can be folded relative to each other. The plurality of hard plates are connected to the two body members **1**. In another embodiment, the foldable member **2** is entirely made of a flexible material. In yet another embodiment, the foldable member **2** is composed of a flexible outer layer **21** and a plurality of hard plates arranged in the flexible outer layer **21**, the flexible outer layer **21** is connected to the two body members **1**, and the foldable member **2** can be folded at a junction between each two adjacent hard plates.

[0028] The heating assembly **3** is distributed on the foldable member **2** and at least one of the two body members **1**. The heating assembly **3** is configured to heat the heating cavity **100**. To further improve heating efficiency of the heating cavity **100**, the heating assembly **3** is distributed as widely as possible over an entirety of the foldable member **2** and the two body members **1**, such that the heating assembly **3** can substantially surround the heating cavity **100**, thereby improving the heating efficiency.

[0029] Additionally, as shown in FIG. **1**, a support structure may be mounted on the body member **1** to lift the two body members **1** and the foldable member **2** off the ground.

[0030] In some embodiments, referring to FIGS. **5-7** and **10-12**, the heating assembly **3** includes a control board **31**, a flexible heating sheet **32**, and at least one heating plate **33**. Each of the flexible heating sheet **32** and the heating plate **33** is electrically connected to the control board **31**. The control board **31** is mounted on the foldable member **2** or either of the two body members **1**. The flexible heating sheet **32** is mounted on the foldable member **2** and is capable of folding or unfolding with the foldable member **2**. The at least one heating plate **33** is mounted on at least one of the two body members **1**. Particularly, each of the two body members **1** is arranged with one heating plate **33**. The control board **31** is a PCB circuit board. Either of the two body members **1** can be arranged with a control panel which is electrically connected to the control board **31**, allowing users to adjust a working state of the heating assembly **3** via the control panel.

[0031] In some embodiments, referring to FIG. **5**, the flexible heating sheet **32** includes a first mounting sheet **321** and a first heating wire **322** distributed on the first mounting sheet **321**. The first mounting sheet **321** is fixedly mounted on the foldable member **2**, such as by sewing or adhesive bonding. The first heating wire **322** is composed of a high-resistivity wire and an insulating outer layer covering the high-resistivity wire. The first mounting sheet **321** is configured to fix a shape of the first heating wire **322**, such as by sewing the first heating wire **322** to the first mounting sheet **321** to prevent displacement of the first heating wire **322**.

[0032] Referring to FIGS. **5** and **6**, a distribution path of the first heating wire **322** is arranged as a continuous curve, primarily to increase a distribution area of the first heating wire **322** on the first mounting sheet **321**, thereby expanding a heating area of the first heating wire **322**.

[0033] In some embodiments, as shown in FIGS. **3-4**, the foldable member **2** includes at least three partitions **20** sequentially connected to each other. For ease of description, this embodiment describes the foldable member **2** as having three partitions **20**, with any two adjacent partitions **20** capable of bending toward each other. One partition **20** forms a bottom wall **1002** corresponding to the heating cavity **100**, and the other two partitions **20** face each other, and each of the other two partitions **20** forms a respective one of two side walls **1001** corresponding to the heating cavity **100**. Each of the two body members **1** forms a respective one of the other two side walls **1001**. The three partitions **20** and the two body members **1** cooperatively define the heating cavity **100** which is prism-shaped. The three partitions **20** and the two body members **1** further cooperatively form an opening at a top of the heating cavity **100** and communicated with the heating cavity **100**. The opening is used for inserting towels. A cover may be added to the opening to open or close the

heating cavity **100**.

[0034] It should be noted that each partition **20** is at least partially flexible, allowing the each partition **20** to fold as the two body members **1** move close to each other and to unfold as the two body members **1** move away from each other.

[0035] The first heating wire **322** is locally densely arranged to form at least three heating portions **3220**. Each of the at least three heating portions **3220** is distributed on a respective one of the at least three partitions **20**. Each heating portion **3220** of the first heating wire **322** is arranged as a continuous curve, such as a continuous “U” shape. For example, in one embodiment shown in FIGS. **3**, **6**, and **7**, the first heating wire **322** has three heating portions **3220**, and each of the three heating portions **3220** is distributed on a respective one of the three partitions **20**. In other words, the three heating portions **3220** are distributed on the bottom wall **1002** and the two side walls **1001** corresponding to the heating cavity **100**. When one heating plate **33** is mounted on each of the two body members **1**, each of the bottom wall **1002** and four side walls **1001** corresponding to the heating cavity **100** can emit heat toward the heating cavity **100**. In another embodiment, the first heating wire **322** has six heating portions **3220**, and each two heating portions **3220** are distributed on a respective one of the three partitions **20** (not shown in the drawings).

[0036] In some embodiments, referring to FIGS. **6-8** and **10**, at least one body member **1** defines a mounting cavity **10**. The control board **31** is mounted in the mounting cavity **10**. The first heating wire **322** is arranged with a connection end **3221** at the bottom wall **1002**, which extends out of the foldable member **2**, enters the mounting cavity **10**, and is electrically connected to the control board **31**.

[0037] Specifically, the body member **1** includes a shell **11** and a mounting frame **12** mounted in the shell **11**. The shell **11** is particularly rigid, and the mounting cavity **10** is defined in the shell **11**. The mounting frame **12** is configured to mount the heating plate **33**, and the heating plate **33** seals the mounting cavity **10**. A first wire hole **112** is defined in a bottom portion **111** of the shell **11**, and the connection end **3221** extends to the mounting cavity **10** through the first wire hole **112**. Alternatively, the first wire hole **112** can be defined in a bottom portion **121** of the mounting frame **12**. Furthermore, the towel heating device further includes a power cord **4**. The control board **31** is mounted in the mounting cavity **10** of one of the two body members **1**. A second wire hole **113** is defined in the bottom portion **111** of the shell **11** of the one of the two body members **1**. The power cord **4** passes through the second wire hole **113** and is electrically connected to the control board **31** and is configured to connect the control board **31** to an external power source. Additionally, the control board **31** is mounted on one of the two body members **1**, and the other of the two body members **1** is arranged with a wire winding structure **114** for winding the power cord **4** when the towel heating device is not in use.

[0038] In some embodiments, referring to FIG. **5**, the foldable member **2** is sheet-shaped. The flexible heating sheet **32** is wrapped in the flexible outer layer **21** of the foldable member **2**. The first mounting sheet **321** of the flexible heating sheet **32** includes a first heat conduction layer **3211** and a thermal insulation layer **3212**. The first heating wire **322** is sandwiched between the thermal insulation layer **3212** and the first heat conduction layer **3211**. The first heating wire **322** is sewn or adhesively fixed to the thermal insulation layer **3212**, while the first heat conduction layer **3211** is adhesively or sewn to the thermal insulation layer **3212**, and the first heat conduction layer **3211** covers the first heating wire **322**. The first heat conduction layer **3211** faces the heating cavity **100**, and the thermal insulation layer **3212** faces away from the heating cavity **100**. Particularly, the first heat conduction layer **3211** is an aluminum foil. The heat generated by the first heating wire **322** is conducted to the aluminum foil, which dissipates heat over a larger area into the heating cavity **100**, improving heating efficiency. The thermal insulation layer **3212** reduces a rate of heat dissipation from the first heating wire **322** to an outside of the towel heating device, helping to maintain a temperature in the heating cavity **100**.

[0039] In some embodiments, referring to FIGS. **10-12**, the heating plate **33** includes a metal plate

333, a second mounting sheet **331**, and a second heating wire **332**. The metal plate **333** is mounted on the body member **1**, the second mounting sheet **331** is fixed on the metal plate **333**, and the second heating wire **332** is distributed on the second mounting sheet **331**. Specifically, the second heating wire **332** is fixed on the second mounting sheet **331** to prevent displacement of the second heating wire **332**, such as by sewing, adhesive bonding, or vacuum encapsulation. Alternatively, the second heating wire **332** is embedded in the second mounting sheet **331** and secured via the vacuum encapsulation. The second heating wire **332** is also composed of the high-resistivity wire and the insulating outer layer wrapping the high-resistivity wire.

[0040] Particularly, a distribution path of the second heating wire **332** is arranged as a continuous curve. A structure of the second heating wire **332** is not shown in the drawings but can be arranged similarly to the heating portions **3220** of the first heating wire **322**, primarily to increase a distribution area of the second heating wire **332** on the second mounting sheet **331**, thereby expanding a heating area of the second heating wire **332**.

[0041] In some embodiments, referring to FIG. **11**, the second mounting sheet **331** and the second heating wire **332** mounted on the second mounting sheet **331** are located on a side of the metal plate **333** facing away from the heating cavity **100**. Heat generated by the second heating wire **332** is conducted through the second mounting sheet **331** to the metal plate **333**, and then dissipated by the metal plate **333** into the heating cavity **100**. This allows the second heating wire **332** to utilize the metal plate **333** with a larger surface area for heat dissipation, increasing the heating area while reducing a localized temperature on the metal plate **333** to prevent users from being scalded by direct accidental contact with the heating wire when reaching into the heating chamber **100**.

Furthermore, a thermal insulation net **122** may be added to the mounting frame **12**. The metal plate **333** is located on a side of the thermal insulation net **122** facing away from the heating cavity **100**, and the thermal insulation net **122** is arranged in a honeycomb structure.

[0042] In some embodiments, referring to FIG. **12**, the second mounting sheet **331** includes a second heat conduction layer **3311** and a third heat conduction layer **3312**. The second heating wire **332** is sandwiched between the second heat conduction layer **3311** and the third heat conduction layer **3312**. The second heat conduction layer **3311** is fixed on the metal plate **333**, such as by adhesive bonding. Each of the second heat conduction layer **3311** and the third heat conduction layer **3312** is an aluminum foil. The second heat conduction layer **3311** and the third heat conduction layer **3312** cooperatively wrap the second heating wire **332** by the vacuum encapsulation. The heat generated by the second heating wire **332** is conducted to the second heat conduction layer **3311** and then to the metal plate **333**. The metal plate **333** is a galvanized steel plate, which has high thermal conductivity and low cost.

[0043] Obviously, the embodiments described above are only a part of the embodiments of the present disclosure, and not all of them. The accompanying drawings give some embodiments of the present disclosure, but do not limit the patentable scope of the disclosure, which may be realized in many different forms. Rather, these embodiments are provided for the purpose of providing a more thorough and comprehensive understanding of the present disclosure. Although the present disclosure has been described in detail with reference to the foregoing embodiments, it is still possible for a person skilled in the art to modify the technical solutions recorded in the foregoing specific embodiments or to make equivalent substitutions for some of the technical features therein. Any equivalent structure made by utilizing the contents of the specification and the accompanying drawings of the present disclosure, directly or indirectly applied in other related technical fields, are all the same within the scope of the patent protection of the present disclosure.

Claims

1. A towel heating device, comprising: two body members, arranged opposite to each other and capable of moving close to or away from each other; a foldable member, connected to the two body

members, wherein the foldable member and the two body members cooperatively define a heating cavity, and the foldable member folds as the two body members move close to each other and unfolds as the two body members move away from each other; and a heating assembly, distributed on the foldable member and at least one of the two body members and configured to heat the heating cavity.

2. The towel heating device according to claim 1, wherein the heating assembly comprises a control board, a flexible heating sheet, and at least one heating plate, each of the flexible heating sheet and the at least one heating plate is electrically connected to the control board, the control board is mounted on the foldable member or either of the two body members, the flexible heating sheet is mounted on the foldable member, and the at least one heating plate is mounted on at least one of the two body members.

3. The towel heating device according to claim 2, wherein the flexible heating sheet comprises a first mounting sheet and a first heating wire distributed on the first mounting sheet, and the first mounting sheet is mounted on the foldable member.

4. The towel heating device according to claim 3, wherein a distribution path of the first heating wire is arranged as a continuous curve.

5. The towel heating device according to claim 3, wherein the foldable member comprises at least three partitions sequentially connected to each other, and two adjacent partitions of the at least three partitions are capable of bending toward each other; the first heating wire comprises at least three heating portions, and each of the at least three heating portions is distributed on a respective one of the at least three partitions.

6. The towel heating device according to claim 5, wherein one of the at least three partitions is configured as a bottom wall corresponding to the heating cavity, the other two of the at least three partitions face each other, and each of the other two of the at least three partitions is configured as a respective one of two side walls corresponding to the heating cavity; at least one of the two body members defines a mounting cavity, the control board is mounted in the mounting cavity, the first heating wire is arranged with a connection end at the bottom wall, and the connection end extends into the mounting cavity and is electrically connected to the control board.

7. The towel heating device according to claim 6, wherein each of the two body members comprises a shell and a mounting frame mounted on the shell, the mounting cavity is defined in the shell, the mounting frame is configured to mount the at least one heating plate, a first wire hole is defined in a bottom portion of the shell or a bottom portion of the mounting frame, and the connection end passes through the first wire hole.

8. The towel heating device according to claim 3, wherein the first mounting sheet comprises a first heat conduction layer and a thermal insulation layer, the first heating wire is arranged between the thermal insulation layer and the first heat conduction layer, the first heat conduction layer faces the heating cavity, and the thermal insulation layer faces away from the heating cavity.

9. The towel heating device according to claim 8, wherein the first heat conduction layer is an aluminum foil.

10. The towel heating device according to claim 2, wherein the foldable member is sheet-shaped, and the flexible heating sheet is wrapped in the foldable member.

11. The towel heating device according to claim 2, wherein the at least one heating plate comprises a second mounting sheet, a second heating wire, and a metal plate, the metal plate is mounted on the two body members, the second mounting sheet is fixed on the metal plate, and the second heating wire is distributed on the second mounting sheet.

12. The towel heating device according to claim 11, wherein a distribution path of the second heating wire is arranged as a continuous curve.

13. The towel heating device according to claim 11, wherein the second mounting sheet is arranged on a side of the metal plate facing away from the heating cavity.

14. The towel heating device according to claim 11, wherein the second mounting sheet comprises

a second heat conduction layer and a third heat conduction layer, the second heating wire is arranged between the second heat conduction layer and the third heat conduction layer, and the second heat conduction layer is fixed on the metal plate.

15. The towel heating device according to claim 14, wherein the second heat conduction layer is adhesively fixed on the metal plate.

16. The towel heating device according to claim 14, wherein each of the second heat conduction layer and the third heat conduction layer is an aluminum foil.

17. The towel heating device according to claim 11, wherein the metal plate is a galvanized steel plate.

18. The towel heating device according to claim 2, further comprising a power cord, wherein each of the two body members comprises a shell and a mounting frame mounted on the shell, the shell defines an mounting cavity, the mounting frame is configured to mount the heating plate, the control board is mounted in the mounting cavity of one of the two body members, a second wire hole is defined in a bottom portion of the shell or a bottom portion of the mounting frame, and the power cord passes through the second wire hole and is electrically connected to the control board.

19. The towel heating device according to claim 18, wherein the control board is mounted on the one of the two body members, the other of the two body members is arranged with a wire winding structure, and the wire winding structure is configured to wind the power cord.

20. The towel heating device according to claim 18, wherein the shell is rigid.
